

AI SOAP ↔ REST Bridge - Test Cases & Sample Data

This document contains interview-ready test cases for validating the AI SOAP-to-REST Bridge. Each test includes sample REST input, SOAP conversion expectations, and expected results.

TC-01: Valid Customer Creation

REST Input:

```
{  
  "customerId":101,  
  "name":"Asta",  
  "phone":"9876543210",  
  "email":"asta@example.com"  
}
```

Expected SOAP:

101
Asta

Expected Result:

HTTP 200, Customer Created

TC-02: Missing Required Field

REST Input:

```
{  
  "name":"Asta"  
}
```

Expected Result:

HTTP 400
Error: customerId is required

TC-03: Invalid Email Format

REST Input:

```
{  
  "customerId":101,  
  "email":"invalid-email"  
}
```

Expected Result:

HTTP 422 Validation Error

TC-04: SOAP Service Timeout

Scenario:

SOAP backend does not respond within configured timeout.

Expected Result:

HTTP 504 Gateway Timeout
Audit log entry created.

TC-05: API Key Authentication

Header:

Authorization: Bearer sb_live_XXXXX

Expected Result:
Authorized request accepted.

TC-06: Invalid API Key

Header:
Authorization: Bearer invalid_key

Expected Result:
HTTP 401 Unauthorized

TC-07: High Volume Load Test

1000 concurrent requests.

Expected Result:
System remains stable.
Success rate >95%.

TC-08: API Auto Mapping Verification

SOAP:
CustomerName

Mapped REST Field:
name

Expected Result:
Mapping generated automatically.

TC-09: SOAP Fault Handling

SOAP Fault Returned:
Invalid Customer

Expected REST Response:
{
 "status": "FAILED",
 "message": "Invalid Customer"
}

TC-10: Audit Logging

Verify:
Request ID
Timestamp
Response Status
Execution Time

Expected Result:
Complete traceability.